



ORTHOFINDER 3

• David Emms, Yi Liu, Laurence Belcher, Jonathan Holmes, Steven Kelly

Phylogenetic Orthology inference

Scalable Orthology inference

High accuracy (QFO + OrthoBench)

Easy downstream analysis

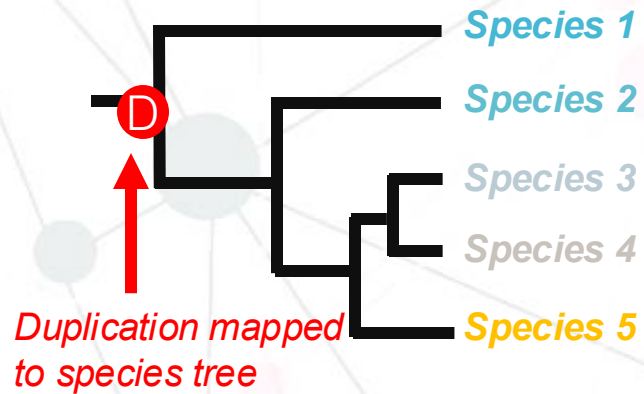
Comprehensive statistics

Duplication identification

ORTHOLOGY INFERENCE

Some useful terms and ideas in Orthology inference: Orthogroup, Ortholog, Hierarchical Orthogroup

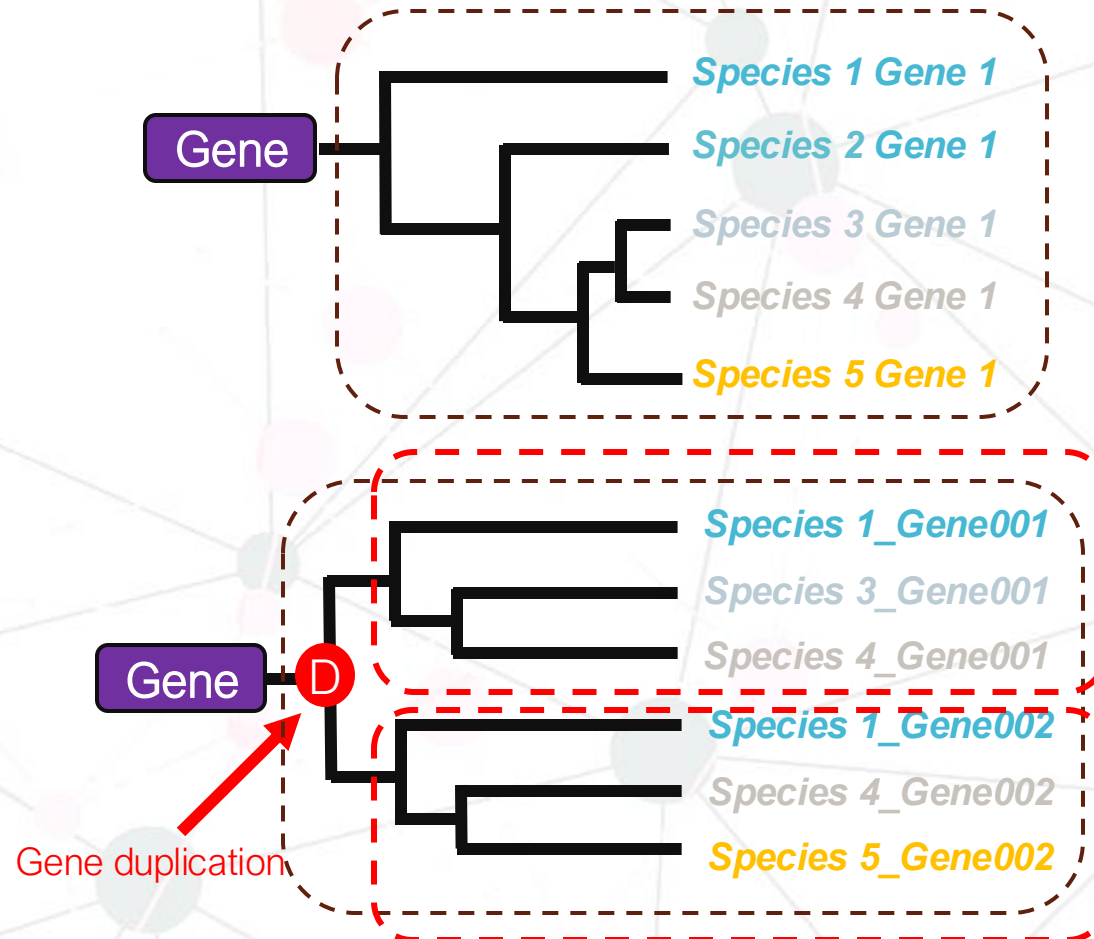
Species Tree



Ortholog: Genes separated by speciation.

Paralogs: Genes separated by gene duplication events.

Gene Tree



Orthogroup: A set of genes descended from a single gene in the last common ancestor of your species.

Hierarchical Orthogroup: A set of genes descended from a single gene at a specific node on a species tree.

ORTHOFINDER

What is OrthoFinder

- OrthoFinder is a fully phylogenetically orthology inference tool.
- We take the approach that orthologs are best identified using phylogenetic trees. We therefore focus on creating and traversing phylogenies rather than graph space.
- OrthoFinder identifies Orthogroups, Hierarchical Orthogroups, Orthologs, gene duplication events.

Current Team:

- Yi Liu
- Laurence Belcher
- Jonathan Holmes
- Steven Kelly

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OrthoFinder : History

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OrthoFinder: phylogenetic orthology inference for comparative genomics

2019. David M. Emms & Steven Kelly

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OrthoFinder: improved phylogenetic orthology inference with enhanced accuracy and scalability

[David M. Emms](#), [Yi Liu](#), [Laurence Belcher](#), [Jonathan Holmes](#) & [Steven Kelly](#) ✉

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OrthoFinder: improved phylogenetic orthology inference with enhanced accuracy and scalability

2026. David M. Emms, Yi Liu, Laurence Belcher, Jonathan Holmes & Steven Kelly

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What is the focus of OrthoFinder

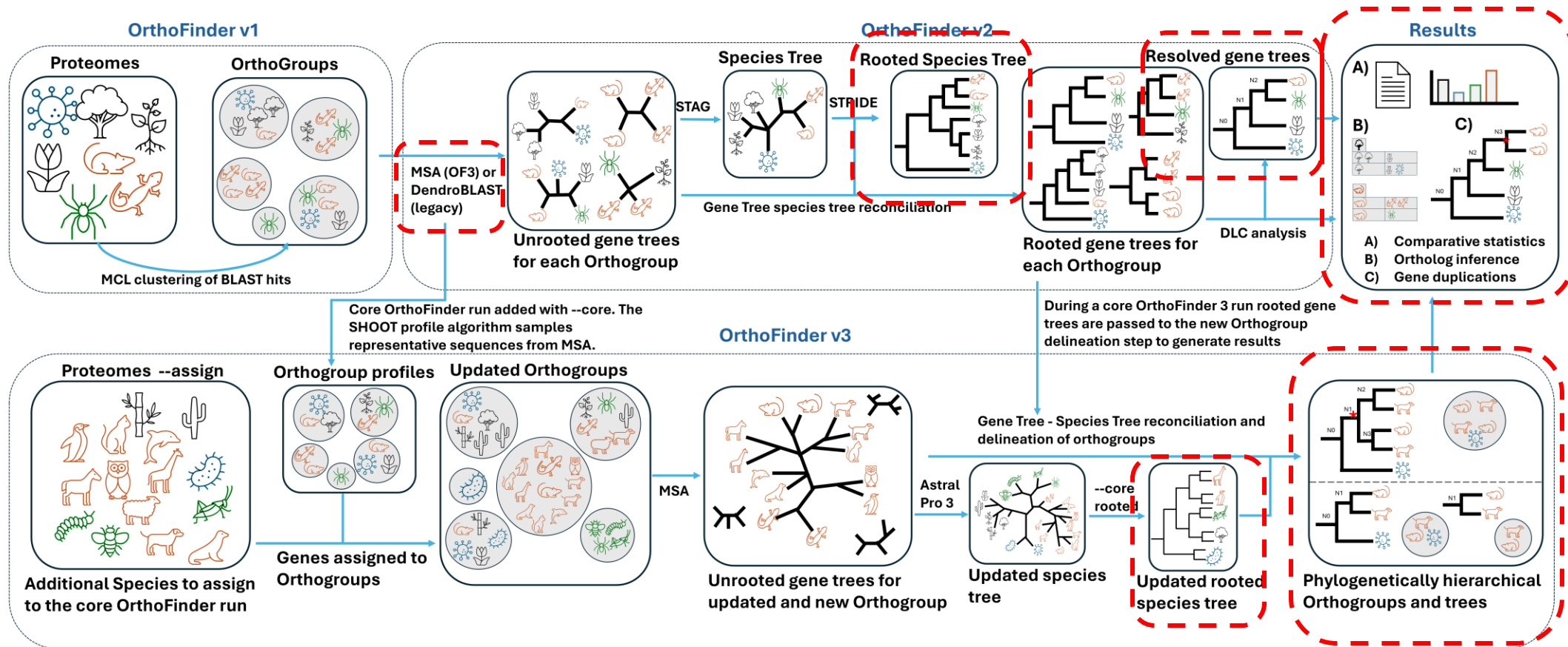
- We aim to produce a tool that is simple to use, freely available and useful to every kind of user
- We want to provide a fully *de-novo* tool it does not require any pre-processing of your data, requiring only a set of proteome files
- We aim to provide a highly accurate method, using cutting-edge benchmarking tools to ensure we provide users with accurate and consistent results.
- OrthoFinder should be able to scale to meet the needs of future researches, with increasing data we must make more scalable methods.
- Provide users with a comprehensive output suite allowing for ease of downstream analysis and allow third party tools to be developed from the results benefiting the whole ecosystem

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The core algorithm of OrthoFinder

OrthoFinder runs with 2 modes. Default and linear. Each have different workflows.

Default runs well on smaller datasets < 64 while linear allows for more rapid scaling up of runtime 64 <.

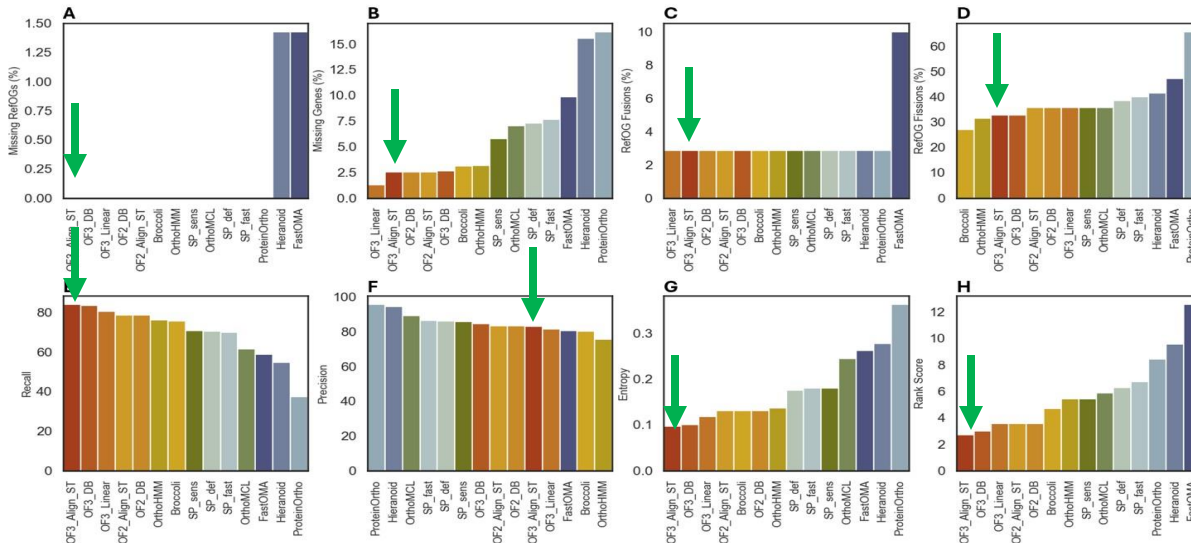


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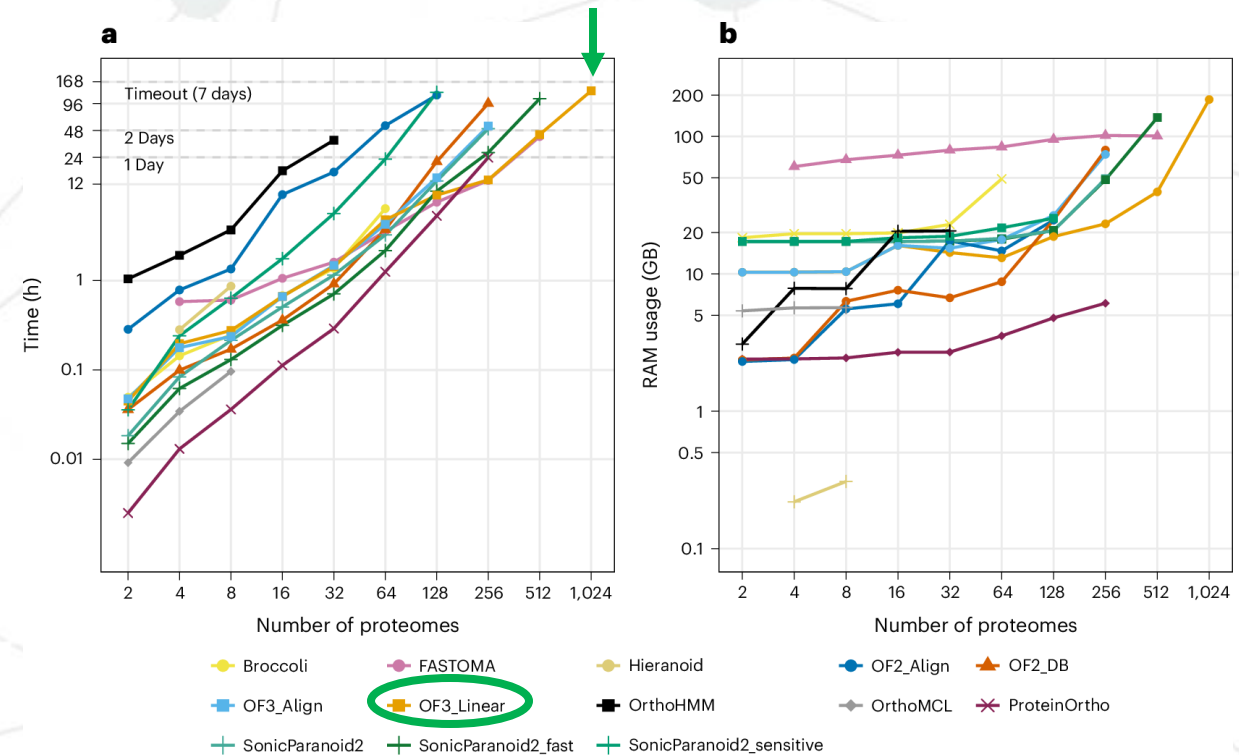
What is OrthoFinder? There's great tools out there why would I use it? How do I run it? Where do I get it?

Why choose OrthoFinder

- OrthoFinder not only infers OGs, HOGs and orthologs but produces comprehensive statistics, phylogenies, multiple sequences alignments, species trees and more!
- High accuracy on benchmark data (OrthoBench + QFO).
- Scalable to thousands of species.
- Fully de novo – only requiring proteome files no pre-processing.



OrthoBench benchmarking (← left is better)



Run time on increasing input size, OrthoFinder can run on 1024 species.

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What is OrthoFinder? There's great tools out there why would I use it? How do I run it? Where do I get it?

Where do I get it and how do I run it

- OrthoFinder can be found on our GitHub page: <https://github.com/OrthoFinder/OrthoFinder>
- Or can be installed through Conda: <https://anaconda.org/bioconda/orthofinder>

OrthoFinder runs with minimal input requiring only a folder of proteome files. It is run through the command line and has a wide range of options.

• Method choices

Parameter	Description	Default	Options
<code>-M</code>	Method for gene tree inference.	<code>msa</code>	<code>dendroblast</code> , <code>msa</code>
<code>-S</code>	Sequence search program	<code>diamond</code>	<code>blast</code> , <code>diamond</code> , <code>diamond_ultra_sens</code> , <code>blastp</code> , <code>mmseqs</code> , <code>blastn</code>
<code>-A</code>	MSA program, requires <code>-M</code>	<code>famsa</code>	<code>famsa</code> , <code>mafft</code> , <code>muscle</code> , <code>msa</code>
<code>-T</code>	Tree inference method, requires <code>-M</code> <code>msa</code>	<code>fasttree</code>	<code>fasttree</code> , <code>fasttree_fastest</code> , <code>raxml</code> , <code>iqtree</code>
<code>-I</code>	MCL inflation parameter	<code>1.2</code>	<code>1-10</code>

• Parallel processing options

Parameter	Description	Default
<code>-t</code>	Number of parallel sequence search threads.	All available
<code>-a</code>	Number of parallel analysis threads.	16 or t/8 (whichever lower)

• Input options

Parameter	Description
<code>-d</code>	Input is DNA sequences.
<code>-s</code>	User-specified rooted species tree.

• Other options

Parameter	Description
<code>-1</code>	Only perform one-way sequence search.
<code>-z</code>	Don't trim MSAs (columns $\geq$ 90% gap, min. alignment length 500).
<code>-y</code>	Split paralogous clades below the root of a HOG into separate HOGs.
<code>-h</code>	Print this help text.
<code>-v</code>	Print version.

• Output options

Parameter	Description
<code>-X</code>	Don't add species names to sequence IDs.
<code>-n <txt></code>	Name to append to the results directory.
<code>-o <txt></code>	Specify a non-default results directory.

• Workflow restart options

Parameter	Description
<code>-b <dir></code>	Start OrthoFinder from pre-computed BLAST results in <code><dir></code> .

• Workflow stopping options

Parameter	Description
<code>-op</code>	Stop after preparing input files for BLAST.

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What is OrthoFinder? There's great tools out there why would I use it? How do I run it? Where do I get it?

How do I run it?

Let's run a default run.

- 1) Decide on your installation method
- 2) For pip create and activate a virtual python environment
- 3) Install OrthoFinder via pip
- 4) Check installation
- 5) Run OrthoFinder using `-f` ProteomeFolder with whatever flags you'd like
- 6) The results will be in ProteomeFolder/OrthoFinder/Results_Data

That's it! OrthoFinder has been executed on the ExampleData Folder.

We can then explore the data as we wish.



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How do I run the linear version

Let's try a more complex linear run Check installation

- 1) Check path to your results for a default "Core" run
- 2) Check path to your proteomes you want to add.
- 3) Run OrthoFinder using --core and --assign.
- 4) The results will be created in the original "core" results directory

The --assign workflow has added our new proteomes to the original Orthogroups, MSAs, phylogenies, statistics are all re-calculated.



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Traversing the results.

- OrthoFinder produces many results files:

```
Results_May13/
├── Citation.txt
├── Comparative_Genomics_Statistics/
│   ├── Duplications_per_Orthogroup.tsv
│   ├── OrthologuesStats_many-to-many.tsv
│   ├── OrthologuesStats_many-to-one.tsv
│   ├── OrthologuesStats_one-to-many.tsv
│   ├── OrthologuesStats_one-to-one.tsv
│   ├── OrthologuesStats_Totals.tsv
│   └── Statistics_Overall.tsv
├── Gene_Duplication_Events/
│   └── Duplications.tsv
├── Log.txt
├── MultipleSequenceAlignments/
│   ├── OG000000000.fa
│   ├── OG000####.fa
│   └── OG000059808.fa
├── Orthogroup_Sequences/
│   ├── OG000000000.fa
│   ├── OG000####.fa
│   └── OG000111601.fa
├── Orthogroups/
│   ├── Orthogroups.GeneCount.tsv
│   ├── Orthogroups.tsv
│   ├── Orthogroups.txt
│   ├── Orthogroups_SingleCopyOrthologues.txt
│   └── Orthogroups_UnassignedGenes.tsv
├── Orthologues/
│   ├── Species_1.tsv
│   └── Species_#.tsv
├── Phylogenetic_Hierarchical_Orthogroups/
│   ├── N0.tsv
│   └── N#.tsv
├── Phylogenetically_Misplaced_Genes/
│   ├── Species_1.tsv
│   └── Species_#.tsv
├── Putative_Xenologs/
│   ├── Species_1.tsv
│   └── Species_#.tsv
├── Resolved_Gene_Trees/
│   └── Resolved_Gene_Trees.txt
├── Single_Copy_Orthologue_Sequences/
│   ├── OG000000000.fa
│   ├── OG000####.fa
│   └── OG000032501.fa
└── Orthogroups_for_concatenated_alignment.txt
```

Run statistics

Multiple sequence alignments and Orthogroup sequences

Orthogroups

Orthologs

HOGs

Resolved Gene Trees

We have a tutorial running through many sections of output on our website:

<https://orthofinder.github.io/OrthoFinder/>

- Beginner tutorials
 - Understanding Orthology
 - Getting input data
 - Advanced Usage
 - Advanced Tutorials
 - Guide to the results
- Yi will be talking you through an example later on

OrthoFinder

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orthogroups, rather than the percentage for all genes across species.

You can see here that we capture most genes across all species.

	<i>Mycoplasma_agalactiae</i>	<i>Mycoplasma_gallisepticum</i>	<i>Mycoplasma_genitalium</i>	<i>Mycoplasma_hyopneumoniae</i>
Number of genes	820	763	476	674
Number of genes in orthogroups	650	596	417	552
Number of unassigned genes	170	167	59	122
Percentage of genes in orthogroups	79.3	78.1	87.6	81.9
Percentage of unassigned genes	20.7	21.9	12.4	18.1

The lowest percentage is the *Mycoplasma_gallisepticum*, but we still managed to assign 78.1% of its genes to orthogroups. The key message here is that it's always a good idea to look at this information before you start interpreting your results. If the numbers were too low for one species, we might want to consider sampling more species to fill in the long evolutionary divergence between species.

One more useful thing to do before we really start to dive in is to look at the species tree. You can do this by opening the tree in [ITOL](#) by either copy and pasting the file content or uploading the file directly.



ORTHOFINDER : A RECAP

OrthoFinder: Who, What, When, Where, Why

- **Who:** The Kelly Lab at the University of Oxford (here today: Yi, Laurence, Jonathan)
- **What:** OrthoFinder is a fast, accurate, reliable and scalable way of doing comparative genomics
- **When:** We make frequent upgrades to the method and are planning many future versions and advances
- **Where:** Find us on Github for the latest releases and branches. Or on Conda for stable releases
- **Why:** We want anyone from bench scientists to HPC users to be able to get the information they need

THANKS!

OrthoFinder

Current Team:

- Yi Liu
- Laurence Belcher
- Jonathan Holmes
- Steven Kelly

Please come ask us any questions or if you have any features you'd like us to implement in the future!

Enjoy the rest of the talks